

Opening:

"Hi, I'm working on a class project at Columbia, exploring how people deal with navigating places without a clear internet connection or GPS. I'm not selling anything. I just want to understand your experience. Would you mind chatting for about 25 minutes?"

Who are they:

"Great. By the way, it's OK to be brutally honest with me. You can't hurt my feelings. That's how we improve our ideas."

Problem exploration:

"Do you run into issues with maps not working inside buildings?"

"Think back to the last time you were inside a large building and tried using your phone to find where you were going. Walk me through what happened."

Current solutions:

"What was the hardest or most frustrating part of that experience?"

"Can you tell me about the last time that happened?"

"How did that make you feel in the moment?"

"How often does that come up?"

Solution signal:

"What have you tried so far to deal with or work around this?"

"What bugs you most about those approaches?"

"Where do those workarounds fall short for you?"

Wrap-up:

"Have you ever spent money trying to solve this—like buying an app, paying for a premium service, or buying a physical tool? If so, how much?"

If no:

- "Got it. What about non-monetary costs—like how much extra time or stress does this usually cost you when it happens?"

If yes:

- "Is there anything else about navigating these kinds of spaces that I didn't ask about, but you think is important?"

Raw Notes:

Opening & Warm-Up

- **Consent & Willingness:** Agreed to chat and help with the class project.
- **General reaction / openness:** Friendly and candid; joked about interview tone before getting into answers.

Problem Exploration

- **Issues with maps indoors (general experience):**
 - Acknowledges blue-dot accuracy and signal loss happen indoors, causing loss of confidence in the map.
 - Does not view indoor navigation as a major painful problem because buildings have inherent physical navigation systems.
- **Walkthrough of last indoor navigation attempt:**
 - Tried finding a restaurant in South Beach / mall environment using phone map.
 - Map failed or was inaccurate → abandoned phone map immediately → looked for physical directory/kiosk instead.

Current Solutions & Frustrations

- **Hardest / most frustrating part of the experience:**
 - Indoor failure isn't very frustrating because alternative solutions (kiosks, signs) are readily available.
 - True frustration occurs when relying on navigation in **outdoor zero-signal areas** (e.g., subways, driving through rural/mountainous areas to soccer matches, hiking on trails) where loss of voice/GPS leads to getting lost.
- **Emotional impact in the moment:**
 - Quickly loses trust in the map app and stops using it.
 - Doesn't get overly emotional or angry indoors—accepts it as a cell coverage limitation and switches modalities.
- **Frequency of the issue:**
 - Indoor maps failing: Occasional, but easily ignored due to alternatives.
 - Outdoor/rural signal loss: Happens when driving in mountains or hiking trails.

Solution Signal & Workarounds

- **Existing workarounds tried:**
 - Physical kiosks and directional directories inside malls.
 - Switching between apps (e.g., trying Apple Maps if Google Maps fails).
 - Relying on room/floor numbering logic (e.g., Room 528 = 5th floor) or asking people nearby.
- **What bugs them most / Where workarounds fall short:**
 - Workarounds work fine inside enclosed spaces like malls.
 - Fall short in wilderness/rural/driving contexts where no kiosks, signs, or people exist to help.

Wrap-Up & Value Signals

- **Financial investment:**
 - Has spent **\$0** on tools/apps to solve indoor navigation.
- **Non-monetary costs:**
 - Minimal personal stress indoors. If a mall lacks clear directory kiosks or navigation, **she simply leaves the mall**—costing the mall business, not her.
- **Additional insights / Unasked important details:**
 - Believes large indoor venues (malls/buildings) can easily solve cell coverage themselves with boosters.
 - Strongly suggested shifting problem focus from **indoor commercial spaces (malls)** to **rural, hiking, driving, or high-stakes environments (e.g., military/defense application)** where GPS/cellular is guaranteed to fail and alternative physical cues are missing

Something to navigate for hiking. Rule areas or mountain areas inclined to think on maps when you cant. Dirving in

1. What You Learned

- **Who:** Jeanmarie a mom of two
- **Problem confirmed?:** No for indoor spaces; Yes for outdoor/rural areas. She dismissed indoor navigation as a non-issue because malls have physical kiosks and structured layout logic. However, she brought up navigating without signal in rural, mountainous, and trail/hiking environments completely unprompted.
- **Key quote:** "I just lost confidence in map and didn't use it. So whenever I'm in a mall... I go and look for other sources... If the mall doesn't have a kiosk where I can figure out where things are, I leave."
- **Emotional intensity:** if 0 if indifferent then 3

2. Solutions and Reaction

- **Current solution:** Indoors, she abandons phone maps to use physical kiosks, asks people, switches between Google and Apple Maps, or relies on numerical floor/room conventions. Outdoors/rural, she wanders around or looks for physical landmarks.
- **Money spent?:** \$0
- **Solution reaction:** Highly skeptical of an indoor map solution because enclosed venues already provide physical directories or could easily install cell boosters if they cared.
- **Biggest concern:** Lacks a strong core need for indoor navigation app alternatives; believes physical surroundings already solve the problem inside buildings

3. Commitments and Signal

- **Referral?:** No explicit referrals offered during the session.
- **Follow-up?:** yes
- **Surprise:** her default response to an indoor map failing isn't frustration, but simply walking over to a kiosk or leaving the venue entirely.
- **Signal:** RED SIGNAL for indoor consumer malls. GREEN SIGNAL for high-stakes offline environments (defense, rural transit, search-and-rescue, or wilderness hiking).

Raw Notes:

Opening & Warm-Up

- **Consent & Willingness:** Agreed to chat and help with the class project.
- **General reaction / openness:** Friendly, cheerful, and enthusiastic throughout the call.

Problem Exploration

- **Issues with maps indoors (general experience):**
 - Confirms running into navigation issues inside buildings.
 - Highlighted two key scenarios: parking garages (trying to load routes before driving out) and massive venues like the American Dream Mall.
- **Walkthrough of last navigation attempts:**
 - **Parking Garage Scenario:** Sitting in the car inside a garage trying to set Apple CarPlay/GPS directions before leaving. Because signal is lost, directions won't load. Forced to exit without knowing which way to turn, creating distraction and needing to pull over immediately outside to put in the address.
 - **American Dream Mall Scenario (August 2026):** Inside the huge mall with family. Dad was trying to pick them up from the outside driving lot/parking structure, but GPS failed for both parties. The group indoors couldn't locate the right exit, and the driver outside couldn't find where they were due to vague signage and lost map signal.

Current Solutions & Frustrations

- **Hardest / most frustrating part of the experience:**
 - Waiting inside the garage in the car while the map refuses to load.
 - Severe friction trying to coordinate a pickup at a massive complex when neither party's map is working.
- **Emotional impact in the moment:**
 - High frustration and distress ("really upset").
 - Added emotional pressure because elderly grandparents were present and unable to walk long distances or wander around uncontrolled environments looking for exits.
- **Frequency of the issue:**
 - Occurs when exiting enclosed parking structures or visiting massive commercial facilities (e.g., American Dream Mall a month prior).

Solution Signal & Workarounds

- **Existing workarounds tried:**
 - Placed a direct phone call to coordinate location and pick-up spot.
- **What bugs them most / Where workarounds fall short:**
 - Phone calls add another layer of friction—verbal communication is unclear when neither person can see what the other is seeing.
 - Driving while shouting over the phone in a busy parking lot/garage is unsafe.

- Workaround completely breaks down if cellular service drops along with GPS.

Wrap-Up & Value Signals

- **Financial investment:**
 - Has spent **\$0** on paid software or apps to solve this (joked about buying a compass).
- **Non-monetary costs:**
 - Lost **30 minutes** of time stranded inside the mall trying to coordinate the exit pickup, causing significant stress.

1. What You Learned

- **Who:** Barnard Senior studying econ and math
- **Problem confirmed?: Yes.** She described major navigation friction unprompted in two distinct indoor/enclosed settings: setting Apple CarPlay directions inside underground parking garages and navigating/coordinating pickups at massive complexes (American Dream Mall).
- **Key quote:** "It's really distracting because then you have to like turn out of the parking garage and find a spot to park so you can put in the address again... it really ruins the flow of your drive."
- **Emotional intensity: 9/10.** Strong emotional distress ("made us all really upset") because elderly grandparents were stranded, unable to walk around long distances while maps were failing.

2. Solutions and Reaction

- **Current solution:** Calling the driver on the phone to coordinate landmark exits verbally, pulling over outside parking garages to reset navigation, or staying put inside the venue.
- **Money spent?:** \$0
- **Solution reaction:** Extremely positive and enthusiastic when the vision/NLP offline navigation agent was pitched ("*This is such an awesome product. I love it. I'm gonna buy it.*").
- **Biggest concern:** Safety and friction during workarounds—yelling over the phone while driving in a busy lot is unsafe, and phone calls fail completely if cellular reception drops alongside GPS.

3. Commitments and Signal

- **Referral?:** None
- **Follow-up?:** Yes
- **Surprise:** The primary pain point inside massive venues isn't just a single person finding a store, but multi-party indoor coordination (e.g., matching a pedestrian at an indoor exit with a driver navigating complex outdoor garage loops).
- **Signal:** GREEN SIGNAL. Strong validation for mega-malls, parking garages, and multi-user coordination where signal loss creates severe delays and emotional friction.